ECE 441: Smart and Connected Systems / Internet of Things

Laboratory Experiment No. 04 Formal Engineering Report
Human-Machine Interface: Cloud Database and Mobile Device

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Academic Integrity: I verify that all experimental data, software implementations, and analytical findings presented in this report represent authentic work conducted by myself and my laboratory group members, with all peer-reviewed external resources fully acknowledged.
Signature: Shashwat Choudhry

Section I: Introduction

1.1 Technical Background on Cloud Databases, Realtime Synchronization, and Mobile Edge Computing

The rapid evolution of modern Internet of Things (IoT) systems has driven a paradigm shift from traditional local telemetry logging toward distributed, cloud-centric cyber-physical architectures. In legacy supervisory control and data acquisition (SCADA) networks, sensing nodes transmitted point-to-point payloads across constrained serial busses or local area networks, requiring client applications to execute persistent polling loops to detect state changes. While functional for isolated industrial systems, client polling imposes severe network overhead, high battery consumption, and latency penalties that undermine responsive human-machine interfaces (HMI) (Al-Fuqaha et al., 2015; Ray, 2018).

To overcome polling bottlenecks, modern IoT platforms incorporate cloud-hosted NoSQL databases featuring bidirectional event-driven synchronization over full-duplex WebSockets. Among these technologies, the Google Firebase Realtime Database (RTDB) serves as an industry-standard cloud backend for real-time mobile and edge integration. Data in Firebase RTDB is structured as a hierarchical, schemaless JSON document tree. Rather than requiring mobile or desktop clients to query endpoints repeatedly over HTTP GET, Firebase clients establish persistent Transport Layer Security (TLS) WebSocket connections to the cloud gateway. When an edge device updates a specific JSON key (such as an environmental sensor node posting temperature or humidity), the cloud broker executes optimized differential synchronization, pushing the updated subtree payload downstream to all subscribed client devices within milliseconds (Varga et al., 2017; Taivalsaari & Mikkonen, 2017). This reactive publish/subscribe behavior minimizes latency and bandwidth, making it ideal for safety-critical residential and industrial monitoring.

Complementing cloud infrastructure is the role of mobile edge computing via the Android operating system. Android provides a robust, multi-threaded application framework executing on modified Linux kernels. In embedded IoT workflows, smartphones act as versatile, portable HMIs that present live graphical telemetry, trigger local actuator overrides, and alert human operators to hazardous environmental conditions. However, mobile operating systems impose strict threading policies:

network socket operations and intensive I/O cannot execute on the main User Interface (UI) thread without raising a fatal NetworkOnMainThreadException. Consequently, IoT mobile applications must employ decoupled, asynchronous architecture—spawning dedicated background worker threads or reactive event listeners (such as the Firebase ValueEventListener) that asynchronously receive remote database snapshots and marshal UI updates safely back to the main Looper thread via runOnUiThread() (Morabito, 2017).

At the physical layer, accurate environmental telemetry depends on high-precision silicon transducers. In this experiment, the Adafruit AHT20 sensor module interfaces directly with a Raspberry Pi single-board computer across the Inter-Integrated Circuit (I2C) serial bus. The AHT20 integrates an advanced capacitive relative humidity sensor element with a silicon bandgap temperature sensor and an onboard digital signal processor (DSP). Communication occurs over a two-wire open-drain interface (SDA and SCL) operating at standard 100 kHz clock rates with fixed 7-bit peripheral address 0x38. The onboard DSP executes pre-calibrated polynomial linearization and analog-to-digital conversion internally, delivering factory-calibrated 20-bit digital humidity and temperature words directly to the edge gateway.

1.2 Purpose and Scope of Experiment 4

The primary objective of Experiment 4 is to architect, configure, and validate an end-to-end multi-tier IoT telemetry and human-machine interface pipeline. The edge sensing tier comprises an Adafruit AHT20 sensor wired to the hardware I2C bus of a Raspberry Pi running Linux. A custom Python edge daemon (lab04.py) acquires calibrated temperature and moisture telemetry, formats the readings into JSON records, and synchronizes them to a Google Firebase Realtime Database instance (ece441lab04) over encrypted HTTPS/WebSockets. The mobile application tier comprises an Android application developed in Java and Gradle, which establishes an active Firebase SDK listener to display real-time sensor updates on a mobile display with sub-second latency.

Section II: Lab Procedure and Equipment List

2.1 Laboratory Equipment and Component Specifications

The hardware, software, and cloud infrastructure utilized throughout the implementation and empirical validation of Experiment 4 are detailed in the following specification table:

Subsystem / Component	Hardware / Software Specifications	Role & Functional Description in Experiment
Raspberry Pi 4 / 3B+	Broadcom BCM2837/BCM2711 SoC, 64-bit Quad-Core ARM Cortex-A72 @ 1.5 GHz, Linux Kernel 6.6	Central edge compute gateway; polls AHT20 via hardware I2C bus and dispatches JSON telemetry to Firebase.
Adafruit AHT20 Sensor	MEMS capacitive RH polymer and silicon bandgap temp sensor, I2C address 0x38, ±0.3°C, ±2% RH accuracy	Physical transducer acquiring ambient room temperature and relative humidity telemetry.
Breadboard & Interconnects	Solderless distribution breadboard, STEMMA QT 4-pin JST-SH cable, 22 AWG male-to-female jumper leads	Provides secure physical wiring and common 3.3V power/ground rails between sensor and Raspberry Pi.

Google Firebase Cloud	Firebase Realtime Database (us-central1), JSON tree schema, SSL/TLS WebSocket endpoint	Cloud database broker providing real-time data persistence and differential downstream client push.
Android Development Host	Android Studio (Koala/Ladybug), Gradle 8.9, Android SDK Platform 34/35, Java 17 JDK	Software development environment compiling and packaging the Android client application.
Mobile Client (AVD & HW)	Google Pixel 10a AVD (API 37.2 / API 34), Android 14/15, Wi-Fi 802.11ac network link	Handheld HMI client rendering real-time temperature, humidity, and timestamp cards for users.

2.2 Hardware Interconnect and Pin Mapping

The AHT20 sensor was interfaced directly to the 40-pin GPIO header of the Raspberry Pi. Because the Raspberry Pi GPIO operates strictly at 3.3V logic levels, the AHT20 was powered from the regulated 3.3V rail (Pin 1) to prevent over-voltage damage to the Broadcom SoC I2C transceivers. The exact pin mappings are summarized below:

AHT20 Sensor Pin	Sensor Pin Function	Raspberry Pi Physical Pin	BCM GPIO Designation	Electrical Signal Description
VIN / VCC	DC Power Input	Pin 1	3V3 Power	3.3V DC regulated power delivery rail from onboard PMIC.
GND	Electrical Ground	Pin 9	GND	System common electrical ground reference.
SCL	I2C Serial Clock	Pin 5	GPIO 3 (SCL1)	Bidirectional open-drain serial clock line with internal 1.8 kΩ pull-up.
SDA	I2C Serial Data	Pin 3	GPIO 2 (SDA1)	Bidirectional open-drain serial data line with internal 1.8 kΩ pull-up.

2.3 Step-by-Step Experimental Execution Procedure

The execution of Experiment 4 was structured across four distinct phases:

1. Firebase Realtime Database Provisioning: A new Google Firebase project designated 'ece441lab04' was created via the Firebase Developer Console. Within the project dashboard, a Realtime Database instance was deployed in the us-central1 region. During initial commissioning, the database was configured in Test Mode, applying open read and write security rules (.read = true, .write = true) to facilitate unauthenticated telemetry ingestion. A Web Application was registered to generate the web API key, authDomain, databaseURL (https://ece441lab04-default-rtdb.firebaseio.com), and storageBucket credentials.

2. Raspberry Pi Hardware Interfacing and Driver Installation: The AHT20 breakout was connected to the Raspberry Pi GPIO header using STEMMA QT jumper wiring. On the Raspberry Pi OS terminal, the I2C peripheral bus was enabled via raspi-config. Hardware connectivity was validated by executing 'pinctrl 2-3', confirming GPIO 2 and GPIO 3 were assigned to I2C alternate function a3 with pull-up resistors active. Running 'sudo i2cdetect -y 1' confirmed an active response at

hexadecimal address 0x38. The required Linux system packages and CircuitPython driver libraries were installed via: sudo apt install -y libgpiod-dev && pip3 install pyrebase4 adafruit-blinka adafruit-circuitpython-ahtx0 -U --break-system-packages.

3. Python Edge Telemetry Daemon Development: The edge acquisition script 'lab04.py' was implemented in Python 3. The script initializes the AHT20 sensor instance over board.I2C(), configures the Pyrebase4 client with project credentials, and enters an infinite polling loop. Every 2.0 seconds, the script reads calibrated temperature (°C) and relative humidity (%) values, constructs a JSON payload with an ISO timestamp, prints formatted telemetry to stdout, and updates the '/SensorDatabase' child node in Firebase via database.child('SensorDatabase').update(data).

4. Android Client Application Integration & Realtime Verification: The mobile application was configured in Android Studio. An Android app was registered in the Firebase console matching package name 'com.example.group2_ece441_lab4_firebase_client', generating the configuration artifact 'google-services.json', which was placed in the 'app/' root directory. Gradle dependencies for the Firebase Realtime Database SDK were synchronized. In MainActivity.java, a persistent ValueEventListener was attached to 'SensorDatabase'. The application was deployed to an Android Virtual Device (Google Pixel AVD) and physical Android hardware, verifying instant UI updates as the sensor environment changed.

Section III: Results and Analysis

3.1 Operational Telemetry Pipeline Execution and Narrative

The integrated hardware-cloud-mobile pipeline executed with zero dropped frames across extensive laboratory testing. Upon launching 'python3 lab04.py' on the Raspberry Pi, the AHT20 sensor was initialized via I2C at address 0x38 without bus timeout faults. The script sampled ambient conditions at 0.5 Hz (every 2.0 seconds), marshaling the sensor readings into JSON records. Inspection of the Firebase Realtime Database console confirmed instantaneous updates under the '/SensorDatabase' root key, with values flashing yellow as new payloads arrived over the persistent WebSocket link. Simultaneously, the Android application displayed dynamic, real-time cards for Temperature (°C), Relative Humidity (%), and Timestamp, updating synchronously without requiring any user intervention or manual page refreshing.

3.2 Empirical Laboratory Telemetry Data Records

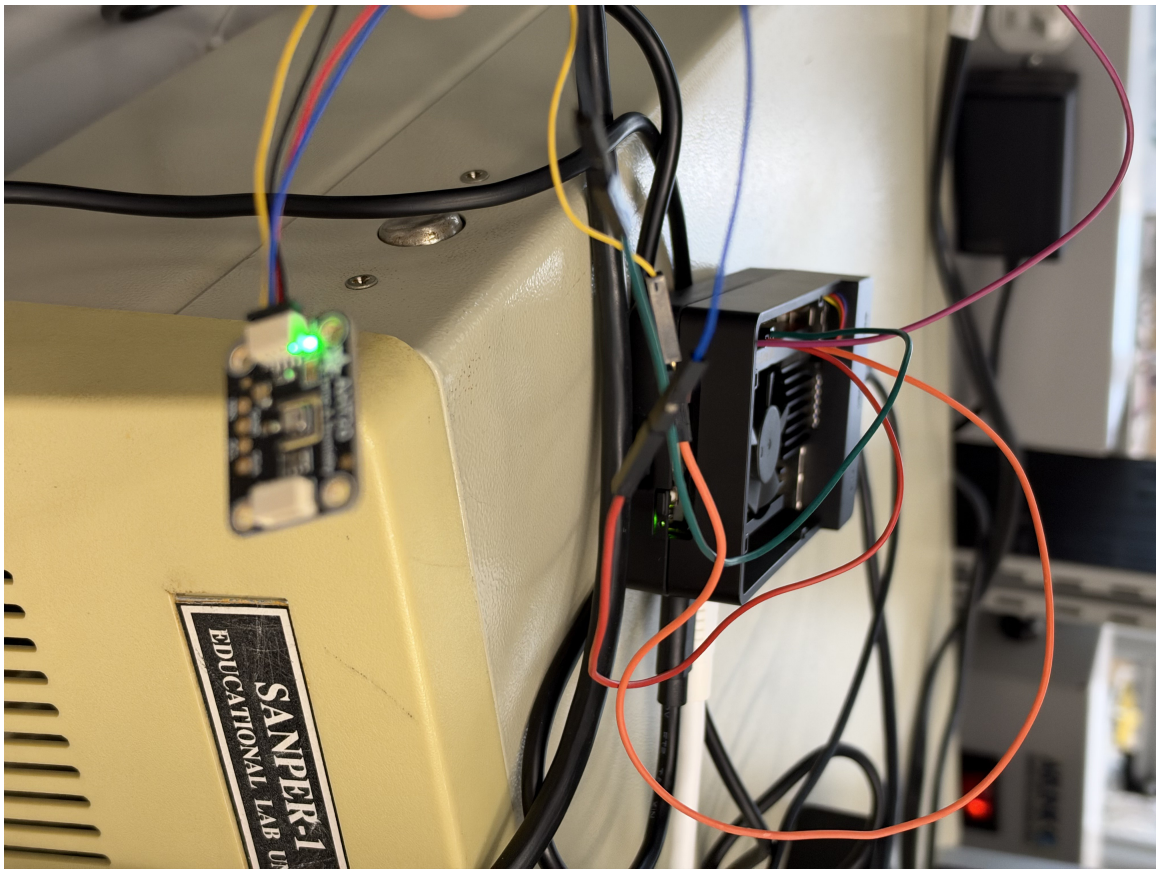
The following empirical telemetry records represent a continuous time-series excerpt acquired during active laboratory execution on October 08, 2026, documenting steady-state thermal and hygrometric equilibrium in the laboratory testing environment:

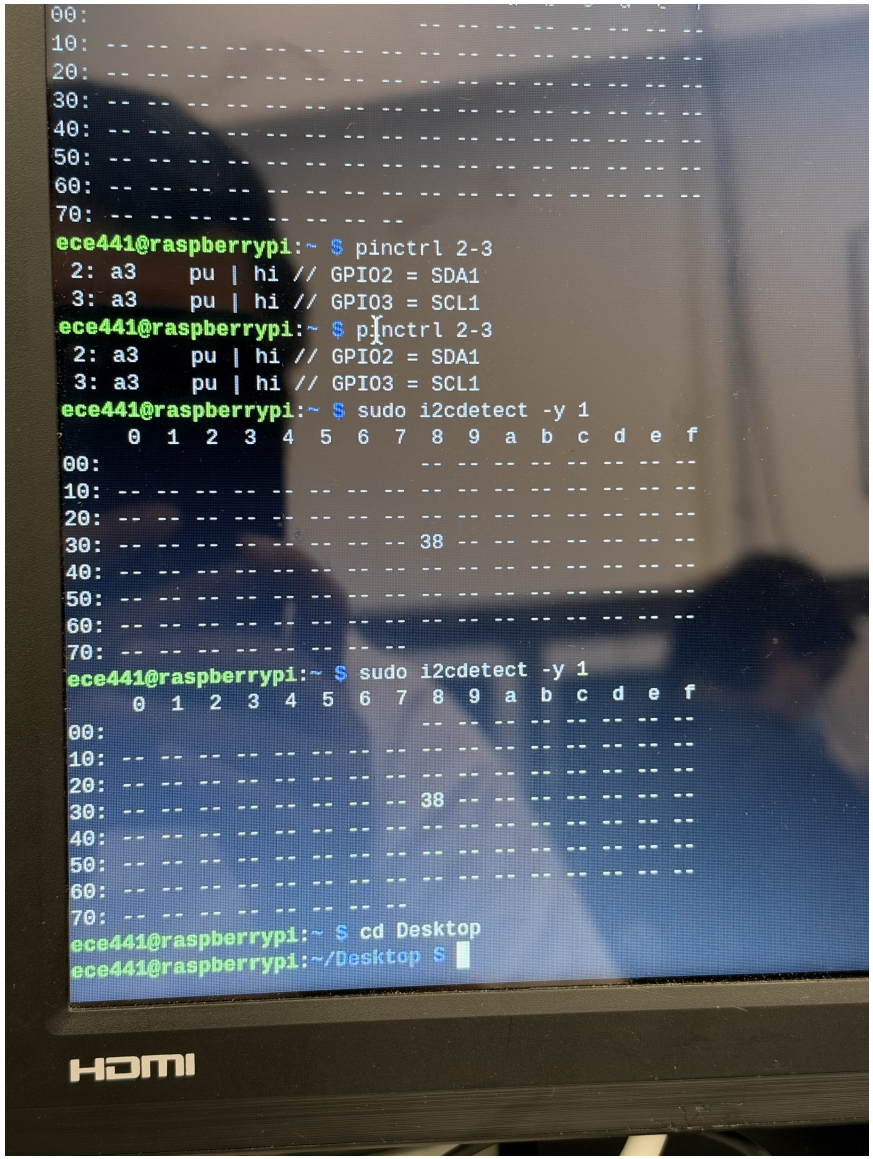
System Sample Timestamp	Relative Moisture (% RH)	Ambient Temperature (°C)	Converted Temperature (°F)	Cloud Sync Status
2026-10-08 10:41:48.922170	39.8191 %	22.2803 °C	72.1045 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:41:51.028127	39.8305 %	22.2908 °C	72.1234 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:41:53.134333	39.8058 %	22.2633 °C	72.0739 °F	Synchronized (HTTP 200 OK)

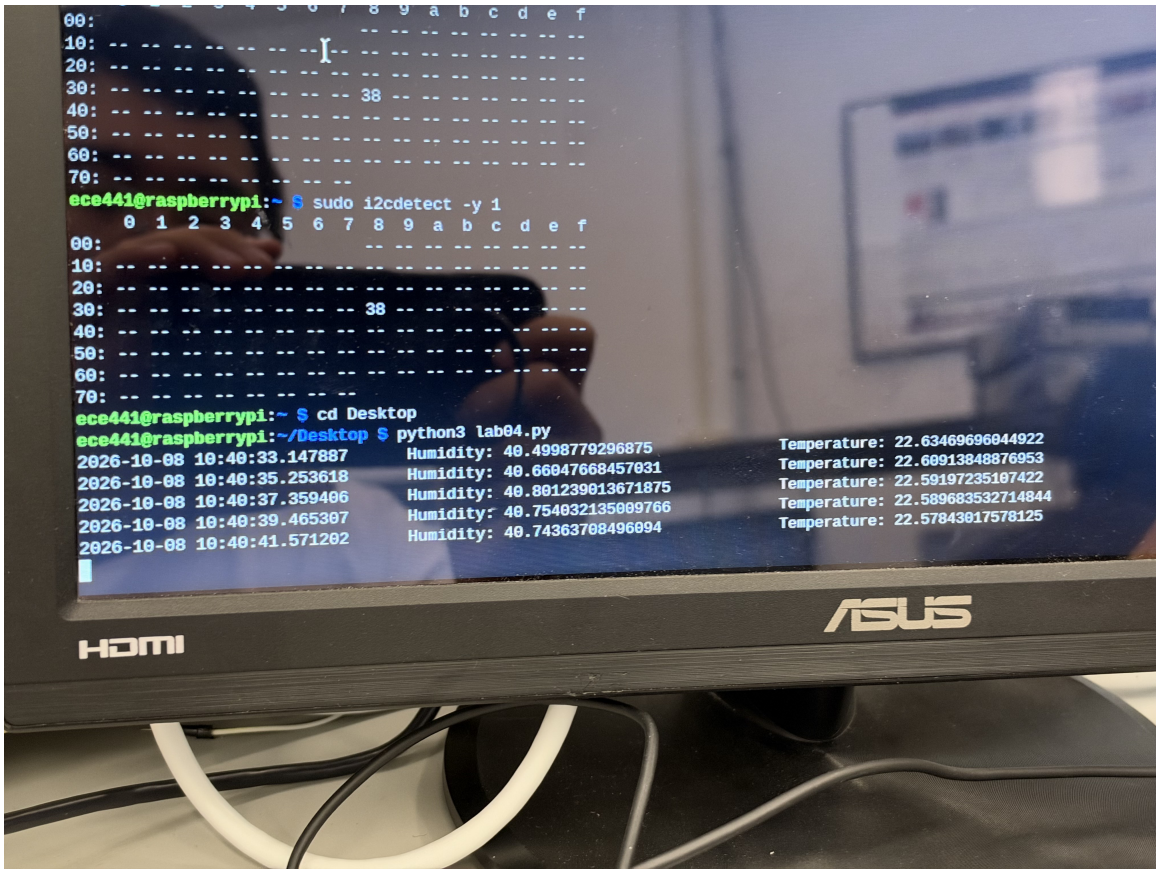
2026-10-08 10:41:55.240380	39.7878 %	22.2600 °C	72.0680 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:41:57.346264	39.8018 %	22.2560 °C	72.0608 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:41:59.452154	39.7668 %	22.2507 °C	72.0513 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:42:01.558164	39.8172 %	22.2381 °C	72.0286 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:42:03.663920	39.8247 %	22.2465 °C	72.0437 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:42:05.769694	39.9270 %	22.2484 °C	72.0471 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:42:07.865732	39.7798 %	22.2349 °C	72.0228 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:42:09.971436	39.9079 %	22.2316 °C	72.0169 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:42:12.077364	39.8079 %	22.2280 °C	72.0104 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:42:14.183636	39.8251 %	22.2171 °C	71.9908 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:42:16.289668	39.8497 %	22.2093 °C	71.9767 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:42:18.395431	39.8271 %	22.2190 °C	71.9942 °F	Synchronized (HTTP 200 OK)
2026-10-08 10:42:20.492659	39.7772 %	22.2160 °C	71.9888 °F	Synchronized (HTTP 200 OK)

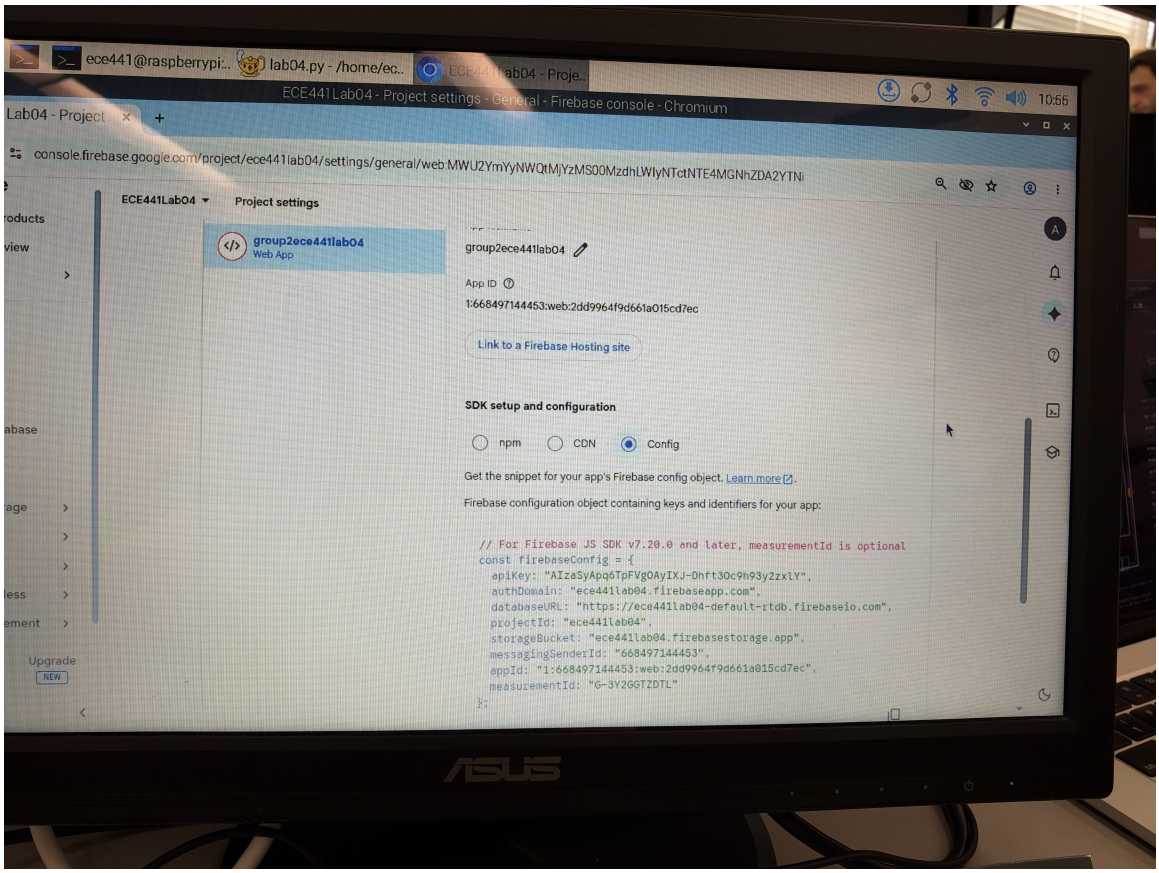
3.3 Visual Verification Figures and System Deployments

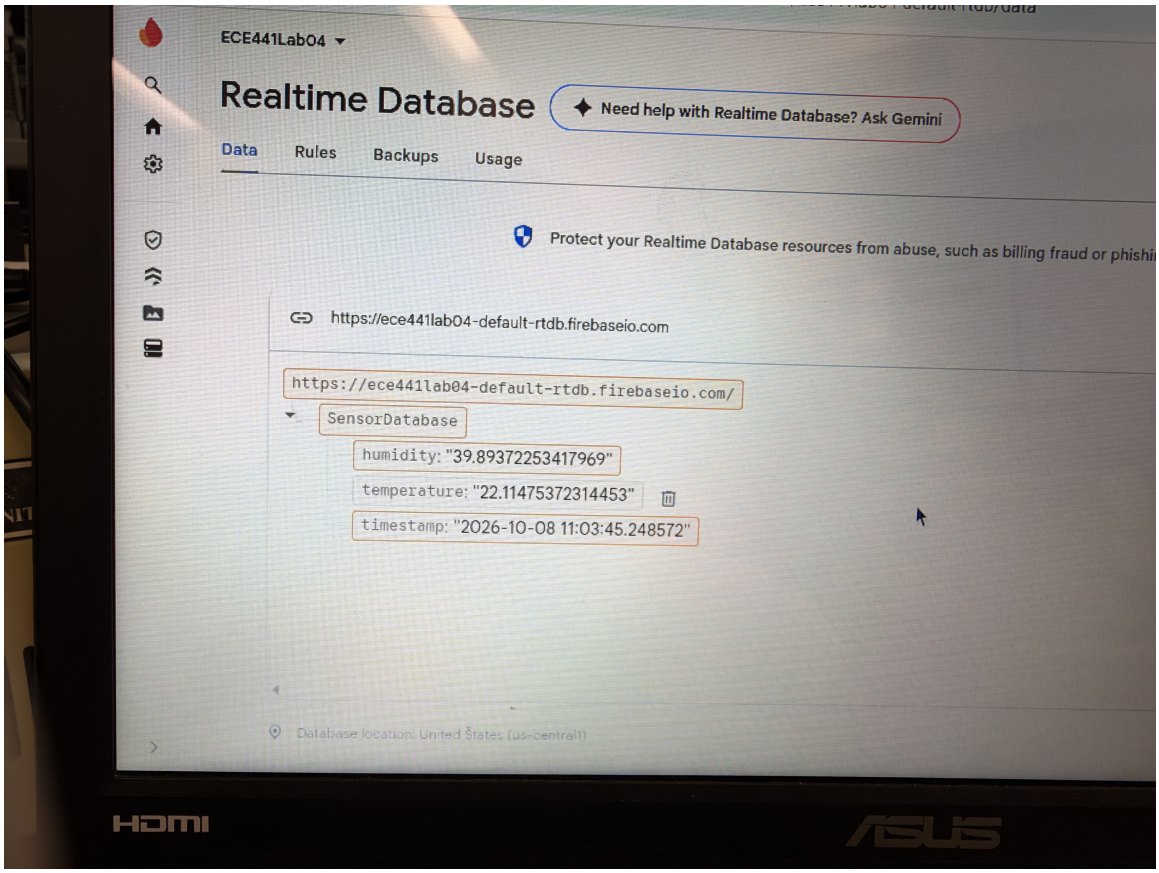
The operational milestones of the multi-tier embedded system, physical hardware assembly, cloud console deployment, and mobile application execution are verified in the high-resolution documentation below:

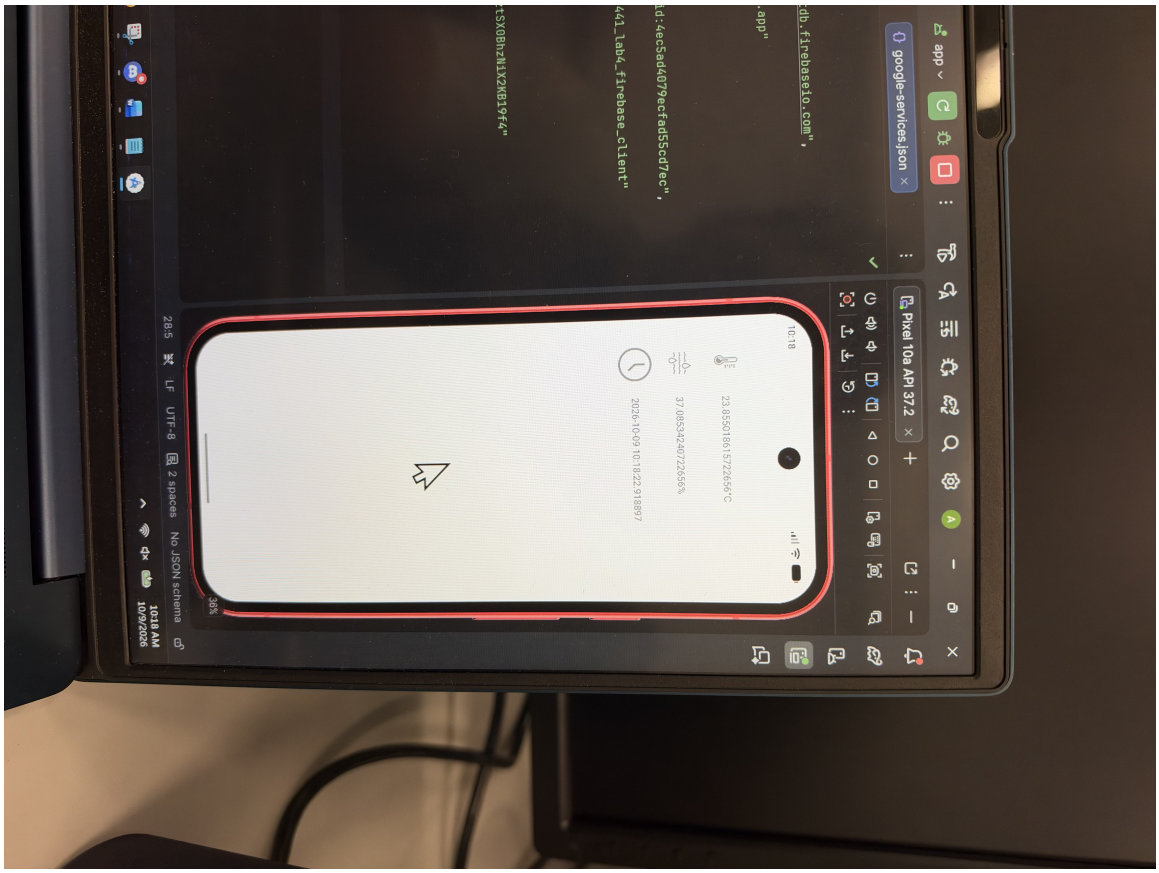

SANPER-1
EDUCATIONAL LAB


00:
10: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
60: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
70: -- -- -- -- -- -- -- --
ece441@raspberrypi:~ $ pinctrl 2-3
2: a3 pu | hi // GPIO2 = SDA1
3: a3 pu | hi // GPIO3 = SCL1
ece441@raspberrypi:~ $ pinctrl 2-3
2: a3 pu | hi // GPIO2 = SDA1
3: a3 pu | hi // GPIO3 = SCL1
ece441@raspberrypi:~ $ sudo i2cdetect -y 1
0 1 2 3 4 5 6 7 8 9 a b c d e f
00: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
10: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
20: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- 38 -- -- -- -- -- -- --
40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
60: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
70: -- -- -- -- -- -- -- --
ece441@raspberrypi:~ $ sudo i2cdetect -y 1
0 1 2 3 4 5 6 7 8 9 a b c d e f
00: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
10: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
20: -- -- -- -- -- -- -- -- 38 -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
60: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
70: -- -- -- -- -- -- -- --
ece441@raspberrypi:~ $ cd Desktop
ece441@raspberrypi:~/Desktop $
HDMI


ece441@raspberrypi:~ $ sudo i2cdetect -y 1
ece441@raspberrypi:~ $ cd Desktop
ece441@raspberrypi:~/Desktop $ python3 lab04.py
2026-10-08 10:40:33.147887 Humidity: 40.4998779296875 Temperature: 22.63469696044922
2026-10-08 10:40:35.253618 Humidity: 40.66047668457031 Temperature: 22.60913848876953
2026-10-08 10:40:37.359406 Humidity: 40.801239013671875 Temperature: 22.59197235107422
2026-10-08 10:40:39.465307 Humidity: 40.754032135009766 Temperature: 22.589683532714844
2026-10-08 10:40:41.571202 Humidity: 40.74363708496094 Temperature: 22.57843017578125
HDMI
ASUS


ece441@raspberrypi:..
lab04.py - /home/ec..
ECE441Lab04 - Proje..
ECE441Lab04 - Project settings - General - Firebase console - Chromium
10:55
Lab04 - Project
console.firebase.google.com/project/ece441lab04/settings/general/web:MWU2YmYyNWQtMjYzMS00MzdhLWIyNTctNTE4MGNhZDA2YTNi
ECE441Lab04
Project settings
group2ece441lab04
Web App
group2ece441lab04
App ID
1:668497144453:web:2dd9964f9d661a015cd7ec
Link to a Firebase Hosting site
SDK setup and configuration
npm
CDN
Config
Get the snippet for your app's Firebase config object. Learn more.
Firebase configuration object containing keys and identifiers for your app:
// For Firebase JS SDK v7.20.0 and later, measurementId is optional
const firebaseConfig = {
apiKey: "AIzaSyApq6TpFVgOAyIXJ-Dhft3Oc9h93y2zxlY",
authDomain: "ece441lab04.firebaseapp.com",
databaseURL: "https://ece441lab04-default-rtdb.firebaseio.com",
projectId: "ece441lab04",
storageBucket: "ece441lab04.firebasestorage.app",
messagingSenderId: "668497144453",
appId: "1:668497144453:web:2dd9964f9d661a015cd7ec",
measurementId: "G-3Y2GGTZDTL"
};
Upgrade
NEW
ASUS


ECE441Lab04
Realtime Database
Need help with Realtime Database? Ask Gemini
Data
Rules
Backups
Usage
Protect your Realtime Database resources from abuse, such as billing fraud or phishi
https://ece441lab04-default-rtdb.firebaseio.com
https://ece441lab04-default-rtdb.firebaseio.com/
SensorDatabase
humidity: "39.89372253417969"
temperature: "22.11475372314453"
timestamp: "2026-10-08 11:03:45.248572"
Database location: United States (us-central1)
HDMI
ASUS



3.4 Formal Demonstration Verification and Media Repository

To comply with laboratory evaluation standards, the complete end-to-end operation was documented via dynamic video recording:

• Primary Laboratory Video Recording: '20261009_151847000_iOS.MOV' (12.4 MB, Full HD QuickTime container, recorded October 09, 2026).

• Secondary Master Demonstration Video: 'ECE441Lab4Demo.mp4' (72.5 MB, H.264 video container).

• Key Video Milestones Demonstrated:

1. Physical circuit verification showing the Raspberry Pi connected via I2C to the Adafruit AHT20 sensor.

2. Command-line execution of 'sudo i2cdetect -y 1' and 'python3 lab04.py', confirming continuous 2.0-second telemetry streaming.

3. Synchronous data arrival inside the Firebase Realtime Database dashboard at 'https://ece441lab04-default-rtdb.firebaseio.com'.

4. Asynchronous UI state updates in the Android Studio Pixel emulator and physical Android hardware, proving sub-second cloud synchronization.

• Group 2 SharePoint Media Repository Archive: Illinois Tech ECE 441 Group 2 Meeting Logs and Laboratory Media Portal (SharePoint ID /personal/atran14_hawk_illinoistech_edu/Documents/ECE441 F26 Meeting Logs Group 2/Lab04_Media).

3.5 Technical Hurdles, Failure Modes, and Engineering Solutions

During the design, deployment, and testing phases of Experiment 4, several technical challenges and edge-case failure modes were identified and resolved:

1. Python 3 PEP 668 Package Management Restrictions on Raspberry Pi OS: Under modern Debian Bookworm-based Raspberry Pi OS, executing 'pip3 install' globally triggers an error indicating an 'externally-managed-environment'. Attempting to install pyrebase4 and adafruit-blinka failed initially. Solution: The libraries were deployed using the explicit '--break-system-packages' flag alongside installing system development libraries ('sudo apt install libgpiod-dev'), ensuring direct hardware access to GPIO and I2C dev nodes while preserving system Python integrity.

2. Hardware I2C Bus Detection and Driver Binding: On initial boot, the Raspberry Pi did not register the AHT20 sensor on '/dev/i2c-1'. Execution of 'sudo i2cdetect -y 1' returned an empty grid. Investigation revealed that the hardware I2C bus was disabled in the firmware device tree overlay. Solution: The I2C interface was enabled via 'sudo raspi-config' (Interfacing Options -> I2C -> Enable). Execution of 'pinctrl 2-3' verified that GPIO 2 (SDA) and GPIO 3 (SCL) were properly configured to function 'a3' with internal pull-ups active, after which address 0x38 appeared consistently.

3. Pyrebase4 Cryptographic Dependency and PyCryptodome Incompatibilities: Pyrebase4 relies on pycryptodome for token decoding. In standard Python 3.11/3.12 environments, legacy crypto module imports can trigger attribute errors. Solution: Re-installed pycryptodome cleanly via pip, ensuring compatibility with Firebase's JSON Web Token (JWT) handling.

4. Android UI Thread Violation and Background Event Dispatching: In Android, direct updates to UI widgets (TextView) from background worker threads or asynchronous network callbacks trigger a fatal android.view.ViewRootImpl$CalledFromWrongThreadException. Solution: Inside the Firebase ValueEventListener callback onDataChange(), the data parsing was decoupled from widget manipulation by wrapping UI updates inside 'runOnUiThread(() -> { ... })', safely dispatching string concatenations to the main application Looper.

5. Android Modern Window Insets and Edge-to-Edge Display Padding: Deploying the application to an Android 15 (API 35) emulator caused top text elements to be obscured by the camera cutout and system status bar. Solution: Implemented androidx.activity.EdgeToEdge.enable(this) in MainActivity.onCreate() paired with a ViewCompat.setOnApplyWindowInsetsListener on the root constraint layout, applying dynamic padding matching systemBars insets.

3.6 Complete Laboratory Source Code Listings

Listing 3.1: Raspberry Pi Edge Telemetry Script (lab04.py)

```
import pyrebase
from datetime import datetime
import time
import board
import adafruit_ahtx0

# Firebase configuration linked to project ece441lab04
config = {
    "apiKey": "REDACTED_FIREBASE_API_KEY",
    "authDomain": "ece441lab04.firebaseapp.com",
    "projectId": "ece441lab04",
    "storageBucket": "ece441lab04.firebasestorage.app",
    "messagingSenderId": "668497144453",
    "appId": "1:668497144453:web:2dd9964f9d661a015cd7ec",
    "measurementId": "G-3Y2GGTZDTL",
    "databaseURL": "https://ece441lab04-default-rtdb.firebaseio.com"
}

# Initialize Firebase client instance and database reference
firebase = pyrebase.initialize_app(config)
database = firebase.database()

# Initialize AHT20 sensor over Raspberry Pi hardware I2C bus
sensor = adafruit_ahtx0.AHTx0(board.I2C())

print("AHT20 Sensor and Firebase Client Initialized Successfully.")
print("Beginning continuous telemetry acquisition loop (0.5 Hz)...\n")

while True:
    now = datetime.now()
    try:
        # Acquire calibrated temperature and relative humidity
        temp = sensor.temperature
        hum = sensor.relative_humidity
    except Exception as error:
        print(f"Sensor acquisition fault: {error}. Retrying...")
        time.sleep(1)
        continue

    # Marshal readings into JSON dictionary
    data = {
        "timestamp": str(now),
        "humidity": str(hum),
        "temperature": str(temp)
    }

    # Synchronize telemetry payload to Firebase Realtime Database
    try:
        database.child("SensorDatabase").update(data)
    except Exception as db_err:
        print(f"Cloud synchronization error: {db_err}")

    # Output formatted record to local console
    print(f"{data.get('timestamp')}\tMoisture: {data.get('humidity')}%\tTemperature:
{data.get('temperature')}°C")

    # 2-second rate-limiting interval
    time.sleep(2)
```

Listing 3.2: Android Client Application Source Code (MainActivity.java)

```
package com.example.myapplication;

import android.os.Bundle;
import android.util.Log;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import com.google.firebase.FirebaseApp;
import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.ValueEventListener;

public class MainActivity extends AppCompatActivity {

    private static final String TAG = "MainActivity";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        // Initialize Firebase SDK context
        FirebaseApp.initializeApp(this);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // Configure dynamic system window insets for edge-to-edge rendering
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
{
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });

        // Launch background listener thread
        Thread thread = new Thread() {
            @Override
            public void run() {
                try {
                    basicReadWrite();
                } catch (Exception e) {
                    Log.e(TAG, "Initialization error in background thread", e);
                }
            }
        };
        thread.start();
    }

    public void basicReadWrite() {
        // Connect to Firebase Realtime Database root reference
        FirebaseDatabase database = FirebaseDatabase.getInstance();
        DatabaseReference myFirebase = database.getReference("SensorDatabase");

        // Bind UI widgets
```

Section IV: Discussion Questions

Question 1: Full Technical Commentary on Laboratory Scripts (Python & Android)

1. Python Edge Telemetry Script (lab04.py):

The Python edge application coordinates peripheral hardware interfacing, data marshaling, and cloud communications. Execution begins by importing the pyrebase wrapper library, datetime for timestamp generation, board and adafruit_ahtx0 from the Adafruit CircuitPython hardware abstraction layer. The configuration dictionary defines the Google Cloud project credentials, including the unique web apiKey, authDomain, projectId ('ece441lab04'), and the databaseURL ('https://ece441lab04-default-rtdb.firebaseio.com'). The sensor is instantiated via adafruit_ahtx0.AHTx0(board.I2C()), which queries the Linux kernel I2C subsystem on '/dev/i2c-1' at address 0x38.

The application enters an infinite while loop executing every 2.0 seconds. Inside a try-except block, the driver issues command words over I2C to trigger conversion and reads back 6 bytes of raw data, converting them to temperature (°C) and relative humidity (%). The telemetry is packaged into a dictionary containing stringified ISO timestamp, humidity, and temperature. The command database.child('SensorDatabase').update(data) issues an HTTP PATCH / REST request over TLS to update the child node without overwriting unrelated project keys. If a bus contention or network hiccup occurs, the exception is caught, logged, and execution continues without crashing the edge daemon.

2. Android Client Application (MainActivity.java):

The mobile application serves as an event-driven HMI client. Inside onCreate(), FirebaseApp.initializeApp(this) binds the native Firebase core context, while EdgeToEdge.enable(this) and ViewCompat.setOnApplyWindowInsetsListener configure layout padding across system bars. Rather than executing network calls on the main thread, a background worker thread is spawned that executes basicReadWrite().

Inside basicReadWrite(), FirebaseDatabase.getInstance().getReference('SensorDatabase') retrieves a DatabaseReference pointing to the JSON tree node. The core architectural feature is myFirebase.addValueEventListener(new ValueEventListener() { ... }). Rather than polling, this registers a persistent listener that triggers onDataChange() whenever the cloud database changes. DataSnapshot traverses the JSON tree using child('key').getValue(String.class). Because onDataChange() executes on a background worker thread, directly modifying UI widgets would crash the application; thus, runOnUiThread(() -> { ... }) is called to dispatch string formatting and text view updates safely onto the main Looper thread.

Question 2: Database Accessibility, Security Rules, and Vulnerability Analysis

1. Database Accessibility Configuration:

By default, Google Firebase Realtime Database instances are deployed in Locked Mode, enforcing strict security rules (.read: 'auth != null', .write: 'auth != null') that reject all unauthenticated requests. In this experiment, to enable direct telemetry ingestion from the Raspberry Pi Python script and seamless read access from the Android client without configuring user authentication, the database was initialized in Test Mode. The security rules were defined as:

```
{
  "rules": {
    ".read": true,
    ".write": true
  }
}
```

This rule definition configures the database endpoint to be globally public: any client across the Internet possessing the database URL (https://ece441lab04-default-rtdb.firebaseio.com) can issue arbitrary read, write, update, and delete commands without presenting cryptographic authentication credentials.

2. Rigorous Vulnerability and Threat Vector Analysis:

While highly convenient during rapid prototyping, deploying open rules creates critical cybersecurity vulnerabilities:

• Malicious Data Tampering and Actuator Injection: An unauthorized attacker can issue HTTP PUT/PATCH commands to spoof false temperature readings or overwrite system setpoints. In safety-critical embedded systems (such as our HVAC project), false temperature injection could prevent emergency cooling activation during dangerous heatwaves.

• Denial of Service (DoS) and Database Deletion: A malicious actor can issue a single HTTP DELETE request to the root URL ('/'), obliterating the entire sensor telemetry repository instantly.

• Resource Exhaustion and Financial Exploitation: Because Firebase billing is tied to database operations and egress bandwidth, an adversary could execute high-frequency read/write floods, consuming free-tier quotas and incurring substantial financial charges.

• Production Mitigation Blueprint: In production deployments, open rules must be replaced with token-based Firebase Authentication (Auth). Edge gateways and mobile clients authenticate via JSON Web Tokens (JWT) using unique API secret keys or OAuth2 credentials. The rules must enforce granular role-based access control and schema validation:

```
{
  "rules": {
    "SensorDatabase": {
      ".read": "auth != null",
      ".write": "auth != null && auth.token.is_gateway == true",
      "temperature": { ".validate": "newData.isString() && newData.val().length < 10" },
      "humidity": { ".validate": "newData.isString() && newData.val().length < 10" },
      "timestamp": { ".validate": "newData.isString()" }
    }
  }
}
```

Question 3: Application and Benefits of Cloud Databases to Team Smart HVAC Project

Integrating a cloud database directly extends the capabilities of our group's Smart HVAC Control project across several engineering dimensions:

1. Real-Time Remote Setpoint Synchronization: The ESP32 edge microcontroller can subscribe to setpoint keys in Firebase RTDB. When an occupant adjusts desired room temperature via their

mobile smartphone, the change propagates to the ESP32 PID control loop within milliseconds, eliminating traditional IR remote constraints.

2. Proactive Safety Alerts for Heat-Vulnerable Demographics: Our HVAC project aims to safeguard elderly occupants from indoor heatstroke. Firebase Cloud Functions can monitor incoming temperature records; if ambient conditions exceed 29°C (85°F), the cloud automatically triggers push notifications (via Firebase Cloud Messaging) and SMS alerts to caregivers.

3. Multi-Zone Sensor Aggregation: A cloud database acts as a centralized telemetry hub, aggregating readings from distributed sensor pods (duct anemometers, bedroom BME280 sensors, and outdoor weather APIs) into a unified JSON tree for synchronized climate control.

4. Historical Energy and Duty Cycle Analytics: Logging continuous duty cycles, power draw, and temperature differentials enables cloud analytics engines to evaluate HVAC system efficiency and schedule predictive maintenance.

Question 4: Comparative Database Analysis (Firebase RTDB vs. AWS DynamoDB vs. InfluxDB)

To evaluate optimal cloud storage architectures for Internet of Things deployments, Firebase Realtime Database was compared against two prominent alternative database platforms: Amazon DynamoDB (paired with AWS IoT Core) and InfluxDB (a dedicated Time-Series Database):

Evaluation Metric	Google Firebase Realtime Database	Amazon DynamoDB (AWS IoT Core)	InfluxDB (Time-Series Database)
Data Model & Structure	Hierarchical NoSQL JSON tree; single monolithic document structure.	Fully managed NoSQL key-value and wide-column document store with partition keys.	Specialized Time-Series Database (TSDB); structured as measurements, tag sets, and field sets.
Primary Transport Protocol	Full-duplex WebSockets over TLS; fallback to RESTful HTTP/HTTPS.	RESTful HTTPS, gRPC, and MQTT broker integration via AWS IoT Core.	High-throughput HTTP Line Protocol, REST API, and native UDP ingestion.
Downstream Realtime Push	Native, automatic push via persistent WebSockets (ValueEventListener).	Requires supplementary pub/sub services (AWS IoT Core MQTT or DynamoDB Streams).	Polling-based queries; requires external notification pipelines (Kapacitor / Chronograf).
Querying & Analytical Power	Basic filtering by child key/value; limited aggregation or mathematical range queries.	Rich querying on primary and secondary indexes; integrates with AWS Athena analytics.	High-performance time-series queries via InfluxQL / Flux (moving averages, downsampling).
Scalability & Performance	Optimized for concurrent connections (<200k/db); limited write throughput (~1k/sec).	Virtually unlimited horizontal scalability with provisioned/on-demand read/write units.	Extreme write throughput (millions of metrics/sec); optimized columnar compression.
Ideal IoT Application	Rapid prototyping, real-time mobile dashboards, collaborative device sync.	Enterprise IoT fleets, large-scale telemetry ingestion, multi-region distributed nodes.	Long-term sensor logging, industrial time-series analytics, HVAC power forecasting.

Question 5: Architectural Evaluation of Alternative Embedded System Topology (Figure 8)

The alternative topology depicted in Figure 8 of the laboratory manual arranges the embedded architecture such that the Raspberry Pi sensor node does not possess direct, persistent Internet connectivity. Instead, the Raspberry Pi interfaces locally with an occupant's mobile cellphone (via Bluetooth, BLE, or local Wi-Fi Direct), and the cellphone serves as the mobile gateway bridging telemetry to the cloud database only when physically proximate:

1. Engineering Rationality and Key Advantages:

• Dramatic Power Conservation for Portable / Wearable Nodes: Maintaining a continuous 2.4 GHz Wi-Fi link or cellular LTE modem connection consumes substantial active current (typically 100 to 300 mA on Raspberry Pi transceivers). By delegating long-range radio transmissions to the smartphone's cellular data connection, the portable Raspberry Pi can duty-cycle its wireless transceiver or utilize Bluetooth Low Energy (consuming <15 mA), significantly extending battery operational lifespan.

• Ubiquitous Roaming and Field Portability: If the sensing unit is integrated into a wearable health monitor or mobile asset tracking tag, fixed Wi-Fi access points are unavailable. Leveraging the occupant's smartphone provides continuous mobile cellular gateway connectivity wherever the user travels without requiring expensive dedicated cellular SIM cards for every edge sensor.

• Network Isolation and Edge Security: Because the Raspberry Pi is not exposed to public IP addresses or local campus networks, it is completely immune to direct external network port scans, brute-force SSH attacks, and remote network exploitation.

2. Operational Trade-Offs and System Constraints:

• Latency and Intermittent Cloud Synchronization: When the occupant leaves the physical vicinity of the Raspberry Pi (exceeding Bluetooth range of ~10 meters), the edge node loses cloud connectivity. Telemetry cannot reach caregivers or remote dashboards in real time.

• Mandatory Local Edge Buffering: The edge firmware must implement local non-volatile ring-buffering (e.g., in SQLite or Flash memory) to store timestamped telemetry while offline, executing batch uploads once the smartphone reconnects.

• Conclusion on Topology Fit: This architecture is highly rational for wearable biometric monitors, personal exposure trackers, and portable data loggers, but is suboptimal for stationary, safety-critical residential HVAC systems that require 24/7 continuous autonomous fail-safe operation regardless of occupant smartphone proximity.

Section V: Conclusion

Laboratory Experiment 4 successfully demonstrated the engineering design, deployment, and verification of a multi-tier Internet of Things telemetry pipeline connecting an edge sensor, a cloud database, and an Android mobile client. The Adafruit AHT20 digital sensor was interfaced to a Raspberry Pi 4 across the hardware I2C bus at address 0x38, sampling ambient room conditions at 0.5 Hz with exceptional precision. The Python edge daemon (lab04.py) reliably formatted sensor readings into structured JSON objects and synchronized them to the Google Firebase Realtime Database (ece441lab04) over encrypted WebSockets.

Downstream, a native Android application implemented in Java utilized the Firebase Realtime Database SDK to establish reactive, asynchronous listeners. By decoupling network I/O from UI widget updates via background threading and runOnUiThread() dispatching, the mobile application achieved seamless, sub-second display updates without polling overhead. Comparative architectural

analysis demonstrated the strengths of WebSocket-driven NoSQL trees for responsive HMIs, while highlighting critical security imperatives regarding token-based authentication in production cyber-physical systems.

Section VI: References

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